

Statistics Roadmap for Machine Learning

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Where this fits: Pipeline map steps 0 (Foundations), 3 (Understand data). See the [Course map](#).

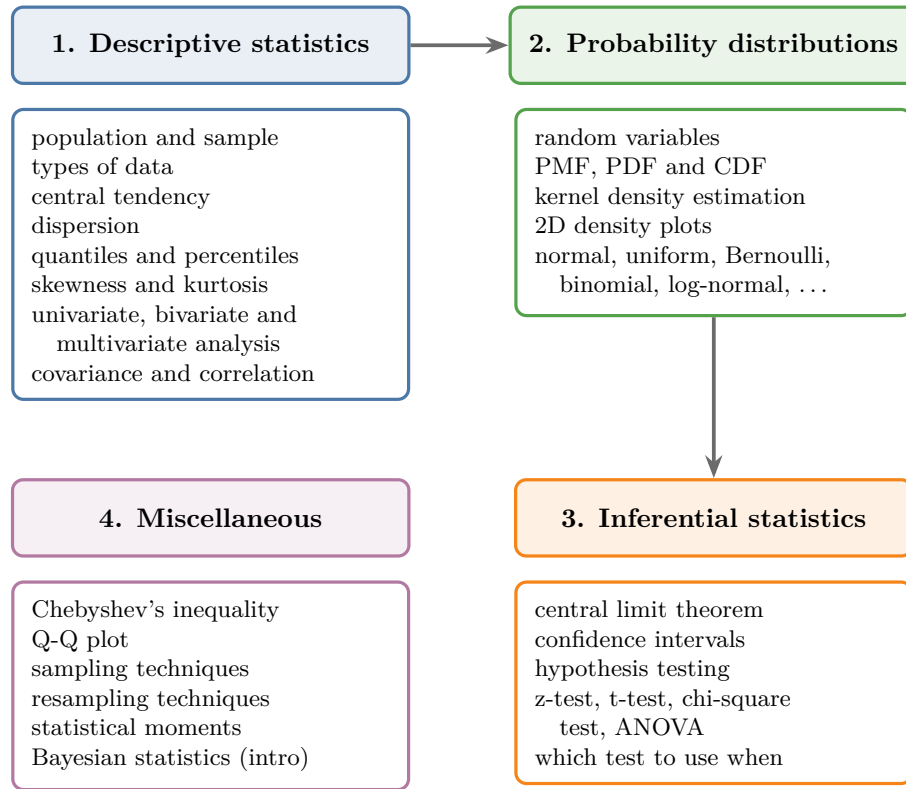


- **Builds on:** Exploratory data analysis ([Note ML-009](#)); Population, sample, parameter and statistic ([Note MA-004](#)); Frequency tables ([Note MA-007](#)); Random variables ([Note MA-020](#)); Confidence intervals ([Note MA-035](#)).
- **Leads to:** Measures of central tendency ([Note MA-005](#)); Variance ([Note MA-006](#)); Percentiles, quartiles and box plots ([Note MA-008](#)); Correlation ([Note MA-009](#)); Normal distribution ([Note MA-020](#)); Probability mass function (PMF) ([Note MA-021](#)).

1. Overview

Key point: Statistics (G-1884) for ML splits into four modules: **descriptive statistics** (G-596), **probability distributions** (G-1571), **inferential statistics** (G-944), and a few miscellaneous tools.

Statistics for ML: four modules



each module builds on the one before it;
the miscellaneous topics are studied when an algorithm needs them

Figure 1: The four modules of statistics for ML

Figure 1 shows the whole map. Each coloured box is one module; the white box under it lists its topics. An arrow means “is needed by”: descriptive statistics comes first, and probability distributions are needed by inferential statistics. Hypothesis testing, in the inferential module, depends on the probability distributions module. This Note says what each module covers, why ML needs it, and where each topic is taught.

2. Why ML needs statistics

Key point: Statistics is how we summarise data and draw conclusions from it, and many ML algorithms are built on it.

Statistics turns raw data into summaries and decisions. In data work it shows up in three places (Figure 2):

- **Understanding data:** every exploratory analysis is built from statistical summaries

and graphs.

- **Algorithms:** linear regression, logistic regression and Naive Bayes all come out of statistics (ISLR ch. 3, §4.3, §4.4.4).
- **Decisions:** checking whether a new medicine, or a new website design, really works better than the old one needs a statistical test (Bruce et al. 2020, ch. 3).

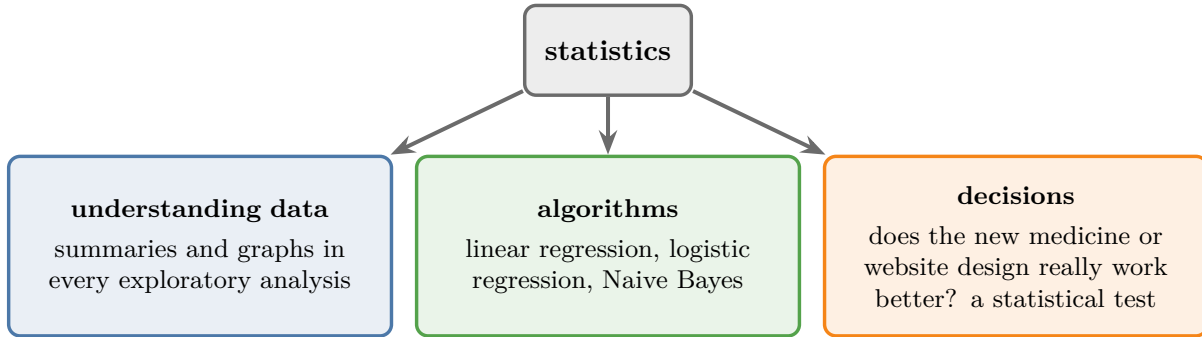


Figure 2: The three places statistics shows up in data work, with an example of each

3. The four modules

Key point: Descriptive statistics summarises the data we have; probability distributions describe the shapes data can take; inferential statistics draws conclusions about data we do not have.

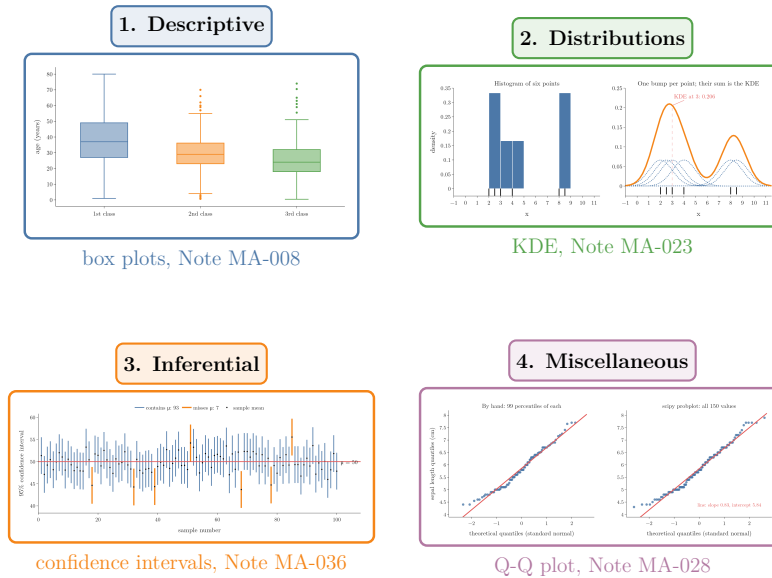


Figure 3: One key figure from each module: box plots of Titanic ages (MA-008), a kernel density estimate built from bumps (MA-023), 100 confidence intervals (MA-036) and a Q-Q plot of iris sepal lengths (MA-028)

Figure 3 previews what each module produces: summaries of the data in hand, a smooth shape for a distribution, a statement about a population with its uncertainty, and a diagnostic plot.

3.1 Descriptive statistics

Key point: Summarise the data we have, with numbers and graphs.

This module describes data that is already in our hands. Its topics, and where each is taught:

Topic	Where it is taught
Population and sample, types of data	population and sample, types of data
Central tendency (mean, median, mode, ...)	what central tendency means
Dispersion (range, variance, standard deviation, ...)	range, variance
Quantiles, percentiles, box plots	percentiles, building a box plot
Skewness	skewness as distance from the normal shape
Kurtosis	kurtosis as tailedness
Univariate analysis	one feature at a time
Bivariate and multivariate analysis	two features: scatter plot, graphs for two features
Covariance and correlation	covariance, correlation

3.2 Probability distributions

Key point: A probability distribution describes how likely each value of a feature is; the hypothesis tests of the next module depend on it.

This module studies the shapes data can take. Here a **feature** (G-772) is an input variable, one column of the data table, and an **observation** (G-1374) is one record, one row of that table:

- **Random variables** (G-1620; a number decided by chance, such as the roll of a die), and the functions that describe them: the PMF (probability mass function, the probability of each count), the PDF (probability density function, the curve for measurements) and the CDF (cumulative distribution function, the running total of probability).
- **Kernel density estimation**, the smooth curve over a histogram (a bar chart of how many values fall in each range), met in [the density plot](#).
- **2D density plots**, the same idea for two features at once.
- **Named distributions**: normal, uniform, Bernoulli, binomial, log-normal and others. The normal distribution's 68-95-99.7 rule appears in [the 68-95-99.7 rule](#).

The basic rules of probability are in [the five basic terms](#) and [the rules every probability follows](#); conditional probability (the probability of one event once another is known) is in [the definition](#).

3.3 Inferential statistics

Key point: Use a sample to say something about the whole population, and say how sure we are.

This module draws conclusions about a **population** (G-1525) from a **sample** (G-1731):

- **Central limit theorem** (G-364): why averages of samples behave predictably.
- **Confidence intervals:** a range built by a method that captures the true population value in a stated share of repeated samples.
- **Hypothesis testing** (G-913): checking a claim about a population with a sample. The common tests are the z-test, t-test, chi-square test and ANOVA (Bruce et al. 2020, ch. 3), and the module ends with when to use which.

3.4 Miscellaneous topics

Key point: A few tools do not fit one module; we study them when an algorithm needs them.

Topic	Where it is taught
Chebyshev's inequality	a later maths Note
Q-Q plot	how a Q-Q plot is built, checking whether a column is normal
Sampling techniques	a later maths Note; sampling bias in sampling noise and sampling bias
Resampling (bootstrap)	bootstrapping
Statistical moments	statistical moments
Bayesian statistics	Bayes' theorem

4. How to study the roadmap

Key point: About 60 hours, or a month at 2 to 2.5 hours a day; some topics can wait until an algorithm needs them.

The whole roadmap takes roughly 60 hours. At 2 to 2.5 hours a day, that is about one month. Two habits help:

- **Know why each topic matters.** For every topic, note where ML or data analysis uses it.
- **Defer some topics.** A few topics, such as Chebyshev's inequality or Bayesian statistics, can wait until an algorithm needs them. The rest are worth covering now, in the order of the map.

Extra: Good resources for this roadmap:

- **Books:** *An Introduction to Statistical Learning* (James, Witten, Hastie, Tibshirani; statistics through ML), *Practical Statistics for Data Scientists* (Bruce, Bruce,

Gedeck; practical, with Python and R), *Think Stats* (Downey; short, Python-based).

- **Online lessons:** Starmer, J., StatQuest, *Statistics Fundamentals*, statquest.org.
- **Revision:** a one-page statistics cheat sheet, once the topics are learned.

5. Summary

Module	Question it answers	Main topics
Descriptive statistics	What does the data we have look like?	central tendency, dispersion, percentiles, graphs, correlation
Probability distributions	What shapes can data take?	PMF, PDF, CDF, normal and other distributions
Inferential statistics	What can a sample tell us about the population?	central limit theorem, confidence intervals, hypothesis tests
Miscellaneous	Extra tools	Q-Q plot, sampling, resampling, moments, Bayes

- Descriptive statistics comes first; inferential statistics depends on probability distributions.
- About 60 hours covers the whole roadmap.

6. Sources

Built from

- CampusX, “Statistics Roadmap for Data Science and Data Analysis | Complete Guide | Full Resources | CampusX”, YouTube, https://www.youtube.com/watch?v=2GV_ouHBw30

Other references

- ISLR: James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021). *An Introduction to Statistical Learning*, 2nd ed. Springer. Chapter 3 (linear regression), sections 4.3 (logistic regression) and 4.4.4 (naive Bayes).
- Bruce, P., Bruce, A. and Gedeck, P. (2020). *Practical Statistics for Data Scientists*, 2nd ed. O’Reilly. Chapter 3, Statistical Experiments and Significance Testing.

7. Key terms

This roadmap adds no terms of its own: each term is defined in the Note that teaches it, and listed in the [glossary](#).